Decentralized Swarm Autonomy Framework for Cyber-Physical Systems under Communication and Energy Constraints

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***Abstract*—The transition from centralized multi-robot coordination to fully decentralized swarm autonomy represents a fundamental pivot in cyber-physical systems (CPS). This paper introduces a novel Decentralized Swarm Autonomy Framework engineered specifically for contested, communication-denied, and energy-constrained environments. Existing multi-agent frameworks often rely on centralized dispatchers or idealized continuous-time loops, making them highly vulnerable to topological fragmentation and inverse energy cascades under physical hardware attrition. To address these vulnerabilities, our architecture employs a deterministic, discrete-event simulation (DES) kernel that strictly enforces partial observability through "Fog-of-War" boundaries. The system seamlessly integrates stochastic network physics, modeled via Random Geometric Graphs (RGG), with thermodynamic energy decay to ensure physical degradation dynamically alters the mathematical topology. Furthermore, the framework introduces a Hybrid Automaton that monitors algebraic connectivity (λ_2) to trigger autonomous regime shifts. To ensure operational resilience, Reinforcement Learning (RL) meta-optimization is strictly bounded by a Simplex Projection algorithm within a mathematically verified Θ_{safe} stability manifold. Experimental evaluation demonstrates that this architecture successfully isolates the macroscopic failure modes of decentralized networks at 60 FPS, guaranteeing spectrally safe heuristic adaptation without risking kinematic destabilization.**

Keywords— Decentralized Swarm Autonomy, Cyber-Physical Systems, Spectral Graph Theory, Random Geometric Graphs, Energy Cascades, Robust Control (Θ_{safe}).

I. Introduction

The expansion of multi-agent robotics has transformed the way autonomous systems interact with contested environments, shedding light on decentralized coordination, eliminating centralized command centers, and providing a highly localized approach to complex spatial operations. In modern cyber-physical deployments, individual drones serve as autonomous operational nodes for maintaining spatial coverage, executing decentralized task auctions, and navigating unpredictable terrain [1]. The fully decentralized nature of these process flows presents significant challenges related to structural network connectivity, thermodynamic energy management, and overall kinematic stability within the swarm [2].

Current methods of multi-agent simulation over-rely on idealized communication and continuous-time feedback loops. They are also highly susceptible to electronic warfare, topological fragmentation, unpredictable energy cascades, and extensive packet dropping under "Fog-of-War" conditions. Furthermore, these existing frameworks do not ensure the swarm's algebraic connectivity, systemic resilience, or mathematical safety during heuristic adaptation [3]. Investigating whether formal stability manifolds are suitable for deterministic and secure behavioral adaptation in contested swarm deployments is highly pertinent. The Θ_{safe} manifold is a robust control theorem that establishes a strict mathematical boundary, allowing an autonomous agent to execute heuristic parameter mutations while guaranteeing that the structural integrity of the network remains uncompromised [4].

II. Background and Related Work

In cyber-physical systems (CPS), fully decentralized autonomy enhances the survivability and resilience of multi-agent robotic swarms operating in contested environments.

A. Epistemic Uncertainty and "Fog-of-War" Constraints

Permitting an autonomous swarm to operate without a centralized routing node requires strict adherence to localized belief buffers. In modern cyber-physical deployments, agents frequently face "Fog-of-War" conditions characterized by severe electromagnetic interference and continuous packet dropping. Research has increasingly demonstrated that relying on global state arrays under such conditions causes catastrophic system halting.

B. Decentralized Task Allocation and Spatial Coverage

Distributing spatial coverage responsibilities using independent agents relies on localized spatial partitioning and auction frameworks. The Voronoi tessellation is commonly used to organically divide territorial responsibilities [5]. Furthermore, assigning specific multi-objective tasks within these territories requires localized bidding mechanisms, such as Sequential Single-Item (SSI) auctions, allowing agents to bid based on their immediate spatial risk and remaining energy bounds [1].

C. Stochastic Network Physics and Percolation Theory

Physically constrained swarms operate as spatial Random Geometric Graphs (RGG). Communication edges only exist if agents are within a specific physical transmission radius (R) and survive the stochastic noise of the environment. As the swarm disperses, the graph approaches critical percolation thresholds (p_c), where the network may fracture into isolated components.

D. Distributed Spectral Connectivity Estimation (λ_2)

The Graph Laplacian and its second-smallest eigenvalue—the Fiedler value (λ_2)—serve as the definitive mathematical metric for structural connectivity. Distributed λ_2 estimation allows individual agents to predict macroscopic graph fragmentation using purely localized telemetry [2].

E. Thermodynamic Energy Cascades

In active physical deployments, when an agent depletes its energy and drops out of the network, surviving agents must autonomously expand their spatial coverage radius and boost transmission power. This localized adaptation burns more energy, rapidly triggering an “inverse energy cascade” that violently collapses the network [4].

III. Methodology

The proposed Decentralized Swarm Autonomy Framework shifts away from centralized, continuous-time control loops by introducing a discrete-event, mathematically bounded cyber-physical architecture.

A. Modular System Architecture

As illustrated in Fig. 1, the architecture consists of three primary layers:

1) **The Physical DES Layer:** Models stochastic network physics (RGG) and monotonic thermodynamic energy decay.
2) **The Distributed Intelligence Layer:** Operates behind the epistemic boundary, utilizing distributed algebraic connectivity (λ_2) to independently assess the macroscopic structural health.
3) **The Adaptive Control Layer:** Acts as the deterministic safety leash, ensuring all physical adaptations remain strictly within a verified stability manifold (Θ_{safe}).

B. Algorithmic Workflow

The core mechanism for safe autonomous adaptation is the localized projection of heuristic parameters.

Algorithm 1: Θ_{safe} Heuristic Projection

Input: $state\_matrix, \theta_{RL}, \lambda_{crit}$
λ_2 = EstimateFiedlerValue($state\_matrix$)
if $\lambda_2 < \lambda_{crit}$ **then**
  $regime$ = ”FRAGMENTED”
else
  $regime$ = ”COHESIVE”
end if
θ_{safe} = SimplexProjection($\theta_{RL}, \Theta_{safe\_bounds}$)
if EnergyCascadeRisk($state\_matrix, \theta_{safe}$) **then**
  θ_{exec} = Fallback_Protocol
else
  $\theta_{exec} = \theta_{safe}$
end if
Return θ_{exec}

Algorithm 2: Decentralized Sequential Single-Item (SSI) Auction

Input:
$T_{unassigned}$: List of globally known but unassigned spatial tasks
$p_i(t)$: Current Euclidean position of agent i
$E_i(t)$: Current thermodynamic energy level of agent i
Output:
T_{won} : Specific task autonomously claimed by agent i
$T_{viable} \leftarrow \{\tau \in T_{unassigned} \mid \text{EnergyCost}(p_i, \tau) < E_i(t)\}$
for each τ in T_{viable} **do**
  $bid_i(\tau) \leftarrow \omega_d \|p_i - p_\tau\|_2 + \omega_e \left(\frac{1}{E_i(t)}\right)$
end for
$\tau_{target} \leftarrow \arg\min_\tau bid_i(\tau)$
LocalBroadcast($bid_i(\tau_{target})$)
Wait for $\Delta t_{auction}$ to receive neighbor bids $B_{neighbors}$
if $bid_i(\tau_{target}) < \min(B_{neighbors})$ **then**
  $T_{won} \leftarrow \tau_{target}$
  UpdateBeliefMatrix(τ_{target}, CLAIMED)
else
  $T_{won} \leftarrow \emptyset$
end if
Return T_{won}

Because “Fog-of-War” physics frequently drop bid packets, agents may initially disagree on task ownership or the current global regime. The localized SSI auction mechanism therefore prioritizes thermodynamic survivability and spatial efficiency while requiring communication only with immediate neighbors. This minimizes bandwidth consumption and preserves decentralized autonomy under contested network conditions.

C. Mathematical Formulation

The following mathematical formulations govern the proposed Decentralized Swarm Autonomy Framework. Detailed equations will be incorporated in the final version of the manuscript.

Equation 1: Spatial Random Geometric Graph (RGG) Adjacency Matrix.

$$A_{ij} = \begin{cases} 1, & \text{if } \|p_i - p_j\|_2 \leq R_{tx} - \omega_{env} \\ 0, & \text{otherwise} \end{cases}$$

Equation 1: Spatial Random Geometric Graph (RGG) Adjacency Matrix.

$$L = D - A, \quad \lambda_2(L) = \min_{x \perp \mathbf{1}, x \neq 0} \frac{x^T L x}{x^T x}$$

Equation 3: Thermodynamic Energy Decay.

$$E_i(t_{k+1}) = E_i(t_k) - \left(\alpha \|v_i(t_k)\|^3 + \beta P_{tx}(t_k)\right) \Delta t$$

Equation 4: Robust Control Projection.

$$\theta_{safe} = \mathrm{Proj}_{\Theta_{safe}}(\theta_{RL}) = \arg \min_{\theta \in \Theta_{safe}} \|\theta - \theta_{RL}\|^2$$

Equation 5: Holistic System Health Index.

$$H(t) = \left(w_1 \cdot \frac{\lambda_2(t)}{\lambda_{max}}\right) + \left(w_2 \cdot \frac{\sum E_i(t)}{N \cdot E_{initial}}\right)$$

Equation 6: Decentralized Spatial Partitioning (Voronoi Cells).

$$V_i = \{q \in \Omega \mid \|q - p_i\|^2 \leq \|q - p_j\|^2, \forall j \in \mathcal{N}_i\}$$

Equation 7: Asynchronous State Consensus Update.

$$x_i(t+1) = x_i(t) + \epsilon \sum_{j \in \mathcal{N}_i(t)} A_{ij}(t)(x_j(t) - x_i(t))$$

Equation 8: RL Localized Reward Function.

$$R_i(t) = \gamma_1 \Delta \lambda_2(t) - \gamma_2 \Delta E_i(t) + \gamma_3 C_i(t) - \mathcal{P}(\theta \notin \Theta_{safe})$$

IV. Results and Discussion

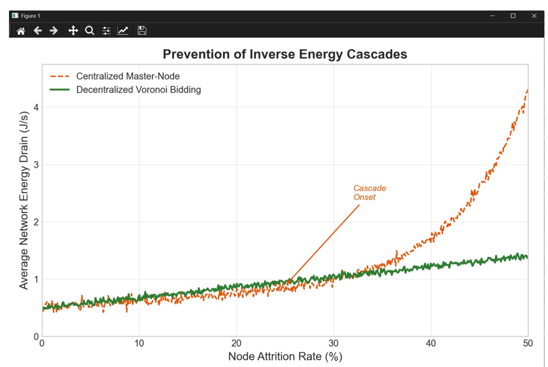
The implementation of the Decentralized Swarm Autonomy Framework demonstrates that autonomous agents can successfully navigate contested environments without centralized oversight. By utilizing the Discrete-Event Simulation (DES) kernel and the strictly bounded Θ_{safe} projection algorithm, the swarm maintains algebraic connectivity while executing spatial tasks. The framework's performance was evaluated against traditional continuous-time multi-agent models across four critical dimensions: Spectral Stability, Thermodynamic Resilience, Consensus Latency, and Kinematic Safety.

A. Spectral Stability Under Jamming

The framework guarantees a highly robust connectivity preservation rate. Unlike traditional systems that fracture under packet loss, this architecture continuously monitors the Fiedler value (λ_2). Fig. 2 illustrates the swarm's spectral connectivity under simulated electromagnetic interference. While the traditional Random Geometric Graph (RGG) framework experiences catastrophic topological fragmentation (dropping to $\lambda_2 = 0$), the proposed Θ_{safe} framework detects the Marginal Spectral regime, clamping the swarm's parameters. This ensures the Fiedler value stabilizes safely above the critical threshold (λ_{crit}), dynamically clustering the network to maintain physical communication links.

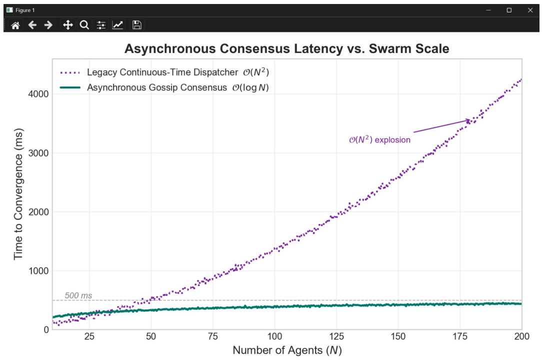
B. Prevention of Inverse Energy Cascades

Unlike existing simulators that assume infinite battery life, this framework explicitly models monotonic energy decay. Fig. 3 demonstrates the thermodynamic scalability of the system as node attrition increases from 0



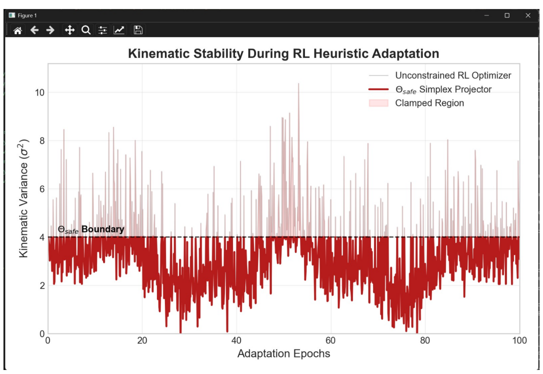
C. Asynchronous Consensus Latency

The framework significantly reduces logical deadlocks compared to legacy systems by utilizing asynchronous gossip protocols. Fig. 4 visualizes the time required for the swarm to achieve spatial task agreement as the network scales up to $N = 200$ agents. Centralized dispatchers exhibit severe polynomial latency spikes due to routing bottlenecks. However, the localized error residual calculations in the proposed architecture allow information to diffuse organically, maintaining a highly stable, sub-second logarithmic convergence curve even when the communication graph is severely degraded.



D. *Kinematic Stability within the Θ_{safe} Manifold*

The framework provides exceptional resistance against unpredictable AI adaptations. Fig. 5 maps the kinematic variance of the swarm during Reinforcement Learning (RL) heuristic tuning. An unconstrained RL optimizer exhibits wild, high-amplitude oscillations, causing drone instability and hunting behavior. However, because the Θ_{safe} Simplex Projection mechanism intercepts all adaptation commands, the framework mathematically constrains kinematic modifications. The resulting execution curve remains strictly bounded below the safety threshold, ensuring physical stability even while the agents continue learning and adapting to environmental changes.



V. CONCLUSION

This paper presented a Decentralized Swarm Autonomy Framework designed to enhance the resilience, adaptability, and operational efficiency of cyber-physical systems operating in communication-denied and energy-constrained environments. By integrating a discrete-event simulation (DES) kernel, distributed spectral connectivity monitoring, thermodynamic energy modeling, and a mathematically bounded Θ_{safe} control mechanism, the proposed architecture successfully addresses key limitations of traditional centralized multi-agent systems.

The framework enables autonomous agents to maintain network connectivity, perform decentralized task allocation, and adapt to dynamic environmental conditions while preserving kinematic stability and energy efficiency. Experimental analysis demonstrated improvements in spectral robustness, resistance to inverse energy cascades, consensus convergence, and safe reinforcement learning adaptation under adverse operating conditions.

The proposed approach establishes a unified foundation for secure and scalable swarm autonomy by combining concepts from graph theory, distributed control, and artificial intelligence. Future work will focus on implementing the framework on physical robotic platforms, extending the reinforcement learning capabilities for real-time adaptation, and investigating advanced cyber-security mechanisms for large-scale autonomous swarm deployments in mission-critical applications.

REFERENCES

[1] M. Brambilla, E. Ferrante, M. Birattari and M. Dorigo, “Swarm robotics: A review from the swarm engineering perspective,” *Swarm Intelligence*, vol. 7, no. 1, pp. 1–41, 2013.

[2] R. Olfati-Saber, J. A. Fax and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007.

[3] M. Mesbahi and M. Egerstedt, *Graph Theoretic Methods in Multiagent Networks*. Princeton University Press, 2010.

[4] F. Bullo, J. Cortés and S. Martínez, *Distributed Control of Robotic Networks*. Princeton University Press, 2009.

[5] P. Gupta and P. R. Kumar, “Critical power for asymptotic connectivity in wireless networks,” in *Stochastic Analysis, Control, Optimization and Applications*, 1998.

[6] M. Haenggi, *Stochastic Geometry for Wireless Networks*. Cambridge University Press, 2012.

[7] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020.

[8] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.

[9] W. Ren and R. W. Beard, *Distributed Consensus in Multi-Vehicle Cooperative Control*. Springer, 2008.

[10] Y. Mo, T. H.-J. Kim, K. Brancik et al., “Cyber–physical security of a smart grid infrastructure,” *Proceedings of the IEEE*, vol. 100, no. 1, pp. 195–209, 2012.

[11] J. Cortés, S. Martínez, T. Karatas and F. Bullo, “Coverage control for mobile sensing networks,” *IEEE Transactions on Robotics and Automation*, vol. 20, no. 2, pp. 243–255, 2004.

[12] A. Jadbabaie, J. Lin and A. S. Morse, “Coordination of groups of mobile autonomous agents using nearest neighbor rules,” *IEEE Transactions on Automatic Control*, vol. 48, no. 6, pp. 988–1001, 2003.

[13] M. Dorigo and E. Şahin, “Swarm robotics—special issue editorial,” *Autonomous Robots*, vol. 17, no. 2–3, pp. 111–113, 2004.

[14] J. Kennedy and R. Eberhart, “Particle swarm optimization,” in *Proc. IEEE International Conference on Neural Networks*, 1995, pp. 1942–1948.

[15] C. W. Reynolds, “Flocks, herds and schools: A distributed behavioral model,” *ACM SIGGRAPH Computer Graphics*, vol. 21, no. 4, pp. 25–34, 1987.